



Pape Magatte Ndiaye DIENE

SOFTWARE ENGINEER | AI & FULL-STACK DEVELOPMENT



Portfolio : <https://magattediene.netlify.app>

CONTACT

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Github

<https://github.com/MagatteDiene>

TECH SKILLS

Languages & Frameworks :

- PHP (Laravel), JavaScript (React, AngularJS), Python (FastAPI, Flask), Java, HTML/CSS, C#

AI / Machine Learning :

- LangChain, RAG, RAGAS, Machine Learning, LLM, Groq API

Databases :

- MySQL, PostgreSQL, ChromaDB

Other:

- API REST/GraphQL, Git/GitHub, Postman, UML Modeling

LANGUAGES

- French
- English



PROFILE

TECHNOLOGY ENGINEER IN COMPUTER SCIENCE, GRADUATE OF THE POLYTECHNIC SCHOOL OF DAKAR (ESP), WITH OVER 2 YEARS OF EXPERIENCE IN WEB APPLICATION DEVELOPMENT, RAG ARCHITECTURES, AND ARTIFICIAL INTELLIGENCE. FULL-STACK DEVELOPER AT HTSOFT SENEGAL, PASSIONATE ABOUT DESIGNING INNOVATIVE SOFTWARE SOLUTIONS WITH STRONG BUSINESS IMPACT.



EDUCATION

TECHNOLOGY ENGINEERING DEGREE, Computer Science 2024 - 2026

Polytechnic School of Dakar (ESP), Dakar

- Year 1: Ranked 1st in class (14.88/20)
- Year 2: Ranked 4th in class (15.05/20)

HIGHER TECHNOLOGY DIPLOMA, Computer Science 2022 - 2024

Polytechnic School of Dakar (ESP), Dakar

Baccalaureate - Science Series (S2) 2021 - 2022

Lycée Sergent Malamine Camara



PROFESSIONAL EXPERIENCE

Fullstack Developer – Backend Focus

HTSOFT SENEGAL

November 2024 - Present

- Designed and developed functional modules for several production business applications (procurement-logistics-distribution ERP, HR information system, real estate management)
- Integrated artificial intelligence solutions into existing applications, including an automated CV screening module
- Contributed to UML modeling and technical architecture of several applications, from functional design to deployment

Web Development Intern

HTSOFT SENEGAL

May 2024 - November 2024

- Designed and built an online seminar management platform from scratch using Laravel, from data modeling to delivery
- Led this project independently as part of the final internship required for the Higher Technology Diploma in Computer Science

CERTIFICATIONS

Elements of AI

University of Helsinki & MinnaLearn

Google AI Professional Certificate

Google

Google AI Essentials Specialization

Google

Introduction to Generative AI for Software Development

DeepLearning.AI

GitHub Foundations

DataCamp

Foundational C# with Microsoft

Microsoft

Cisco Networking Basics

Cisco Networking Academy

Cisco Networking Devices and Initial Configuration

Cisco Networking Academy

OTHER TRAINING

Intensive English Session (Level 3 to Level 4)

CPLA — Centre de Perfectionnement en Langue Anglaise

July 1st, 2026
September 30th, 2026

PROJECTS

SamaVoie — Thesis Project

Hybrid RAG system for academic guidance in Senegal: dense retrieval (BGE-M3) + BM25, RRF fusion, cross-encoder reranking, HyDE and multi-turn query reformulation. Conversational assistant (Kali) + study program recommendation engine.

Technologies : Python, LangChain, ChromaDB, FastAPI, Groq API, React

GMAO — Computerized Maintenance Management System

Full CMMS for two entities of a Senegalese civil engineering and earthworks group, managing a fleet of heavy equipment (machine → sub-assembly → part). End-to-end design (requirements analysis, functional prototype validated with the client, then production application): equipment registry with idempotent Excel import, daily usage tracking with validation workflow, work request → work order (WR → WO) pipeline (Gantt, PERT), preventive maintenance with automatic work order generation, subcontracting synchronized with the STT platform, budget tracking, reliability indicators (MTBF, MTTR, FMECA) and granular roles/permissions.

Technologies : React, TypeScript, Laravel, PostgreSQL, GraphQL

Used by : CSE | SOSETER

SALAMA (Procurement, Logistics and Distribution ERP)

Web application managing the full procurement-transport-distribution cycle for a company: tenders and supplier evaluation, transport offer validation workflow, fleet management (vehicles/drivers/shipments by tonnage), warehouses and delivery notes, sales point network by zone/province, and tracking of donor-funded programs. Integrates Sage X3, a GraphQL API, Excel/PDF exports and a roles/permissions system.

Technologies : Laravel, React, Postman, PostgreSQL

Used by : SALAMA (Madagascar)

SIRH (HR Information System)

Front-end and back-end development of the HR platform, integration of an AI module for CV pre-screening, Laravel API development and application deployment. Ran training sessions for users and supported them in adopting the tool.

Technologies : React, Laravel, Postman, PostgreSQL, OpenAI API

Used by : CSE | FABRIMETAL

GESTIMMO (Real Estate Management ERP)

Rental and financial management software for real estate groups: development of real estate and financial management modules, PDF document generation, data migration. Ran training sessions for clients and supported users.

Technologies : Laravel, Blade, AngularJS, Postman, PostgreSQL

Used by : SERTEM, CMH Sarl, BB Immobilier

N-BaloT — Network Intrusion Classification (Machine Learning)

IoT intrusion detection system based on machine learning, with experimentation on several algorithms (Random Forest, SVM, KNN) and deployment of a web interface.

Technologies : Python (Flask)